

DL Graded Quiz M3

Question 1

In the Gramian Angular Field transformation, the element of the Gram-like matrix in row 1 and column 2 will be the result of what?

- ☐ ϕ_{12}
- ☐ $\cos(x_1 + x_2)$
- ☐ $\cos(\phi_1 + \phi_2)$
- ☒ $\cos(\tilde{x}_1 + \tilde{x}_2)$

Question 2

The main difference between ResNet and DenseNet convolutional neural network architectures is which of the following?

- ☐ ResNet uses 1x1 convolutions while DenseNet does not.
- ☐ ResNet is a multi-branch network while DenseNet features a single branch.
- ☐ DenseNet adds the outputs from the residual and the identity mapping while ResNet concatenates the results.
- ☒ ResNet adds the outputs from the residual and the identity mapping while DenseNet concatenates the results.

Question 3

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TOP

Consider a time series with 5,000 observations. Suppose we implement the GAF transformation to the

data with sequences of 10 observations. By implementing GAF, what will we obtain?

- ☐ 2,500 images of 100x100 pixels
- ☐ 5,000 images of 50x50 pixels
- ☐ 4,990 images of 10x10 pixels
- ☒ 1,500 images of 40x40 pixels

Question 4

Determine which of the following is part of a transfer learning methodology to train convolutional networks.

- ☐ Predicting the labels of a target dataset with a neural network that uses as inputs and training sample the information from a dataset that is different from the target
- ☐ Comparing the performance of a trained neural network in a target dataset with the performance of the same network and the same trained parameters with a dataset different from our target
- ☐ Copying the structure and parameter values of the pre-trained network, except the output layers
- ☒ Comparing the performance of a trained neural network in a target dataset with the performance of the same network trained with a dataset different from the target

Question 5

Determine the correct answer regarding the Gramian Angular Field transformation method.

- ☐ It generates RGB images that determine whether there has been a reduction or increase in the value of the time series.

- ☐ It generates RGB images that contain the polar coordinates of each observation.
- ☐ It constructs images from time series by extracting the linear correlation between observations.
- ☒ It transforms the data by using the polar coordinates of a re-scaled version of the time series.

Question 6

Which of the following convolutional neural network architectures includes multi-branch blocks of layers?

- ☐ NiN
- ☐ ResNet
- ☐ VGG
- ☒ LeNet

Question 7

Consider a time series with 5,000 observations. Suppose we implement the GAF transformation to the data with sequences of 50 observations. By implementing GAF, what will we obtain?

- ☐ 1,000 images of 50x50 pixels
- ☐ 1,000 images of 5x5 pixels
- ☐ 5,000 images of 100x100 pixels
- ☒ 4,950 images of 50x50 pixels

Question 8

Complete the stem:

The Xception architecture for convolutional neural networks _____.

- ☐ represents a useful option when we have at our disposal relatively few examples, compared to a similar architecture that employs standard convolutions
- ☐ concatenates the results of the identity and non-linear mappings in each layer
- ☐ generalizes the VGG blocks to increase the width of the network
- ☒ exploits 1x1 convolutions to get rid of the fully connected layers

Question 9

Determine the correct answer regarding the Gramian Angular Field transformation method.

- ☐ It constructs images from time series by extracting the linear correlation between observations.
- ☐ It generates RGB images that contain the polar coordinates of each observation.
- ☐ It generates RGB images that determine if there has been a reduction or increase in the value of the time series.
- ☒ It requires a rescaling of the time series between -1 and 1.

Question 10

The NiN architecture includes 1x1 convolutions within layers to do which of the following?

- ☐ Reduce the computational cost of the fully connected layers
- ☐ Increase the depth of the layer

- ☐ Increase the number of channels of each output layer
- ☒ Manage input images with large resolution

Question 11

Consider a time series with 1,500 observations. Suppose we implement the GAF transformation to the full data with sequences of 50 observations. Then, what would happen?

- ☐ We would obtain 3,500 RGB images with 50x50 pixels that we can use in a convolutional neural network.
- ☐ We would obtain 1,000 images with 125x125 pixels and a single channel that we can use in a convolutional neural network.
- ☐ We would obtain 1,450 images with 50x50 pixels and a single channel that we can use in a convolutional neural network.
- ☒ We cannot use GAF for sequences of 50 observations.

Question 12

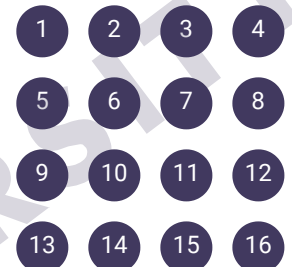
Transfer learning is a useful methodology in which scenario?

- ☐ We have at our disposal sufficient computation resources to train large models.
- ☐ We can only train our model with images that have one channel.
- ☐ We have at our disposal a large amount of training examples to train a deep convolutional neural network.
- ☒ We face a limited number of examples to train a deep convolutional neural network.



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QUESTIONS



Question 13

Once we perform a GAF transformation to a time series, a diagonal element will contain a low value in which of the following cases?

- ☐ The realization is far from the previous realization of the time series.
- ☐ The realization is far from the sample mean of the time series.
- ☐ The realization is close to the sample mean of the time series.
- ☒ The realization is close to the previous realization of the time series.

Question 14

The Inception blocks in convolutional neural network architectures are meant to do which of the following?

- ☐ Recognize different details of the input through filters of different sizes and different parameters
- ☐ Identify an identity mapping

- ☐ Reduce the computational cost of fully connected layers
- ☒ Increase the width and reduce the depth of the network

Question 15

Consider a time series with 5,000 observations. Suppose we implement the GAF transformation to the data with sequences of 500 observations. By implementing GAF, what will we obtain?

- ☐ 5,000 images of 25x25 pixels
- ☐ 1,000 images of 50x50 pixels
- ☐ 4,500 images of 500x500 pixels
- ☒ 500 images of 100x100 pixels

Question 16

Consider the following input image:

$$\begin{pmatrix} 1 & 2 & 4 \\ 3 & 7 & 2 \\ 2 & 0 & 2 \end{pmatrix}; \begin{pmatrix} 0 & 2 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & 0 \end{pmatrix}; \begin{pmatrix} 4 & 2 & 1 \\ 5 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Consider the following 2x2 kernels:

$$\begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}; \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}; \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$$

Consider the following 1x1 convolution kernels:

$$(0 \ 1 \ 0); (2 \ 1 \ 0); (0 \ 1 \ 2)$$

Determine which of the following is the outcome of the separable convolution operator with three output channels.

- ☐ $\begin{pmatrix} 10 & 13 \\ 17 & 8 \end{pmatrix}; \begin{pmatrix} 50 & 73 \\ 71 & 62 \end{pmatrix}; \begin{pmatrix} 28 & 17 \\ 21 & 12 \end{pmatrix}$

☐ $\begin{pmatrix} 39 & 45 \\ 46 & 37 \end{pmatrix}; \begin{pmatrix} 19 & 19 \\ 12 & 21 \end{pmatrix}; \begin{pmatrix} 21 & 41 \\ 42 & 33 \end{pmatrix}$

☐ $\begin{pmatrix} 30 & 43 \\ 44 & 35 \end{pmatrix}; \begin{pmatrix} 29 & 32 \\ 29 & 29 \end{pmatrix}; \begin{pmatrix} 20 & 30 \\ 27 & 27 \end{pmatrix}$

☒ $\begin{pmatrix} 10 & 13 \\ 17 & 8 \end{pmatrix}; \begin{pmatrix} 9 & 2 \\ 2 & 2 \end{pmatrix}; \begin{pmatrix} 20 & 30 \\ 27 & 27 \end{pmatrix}$

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